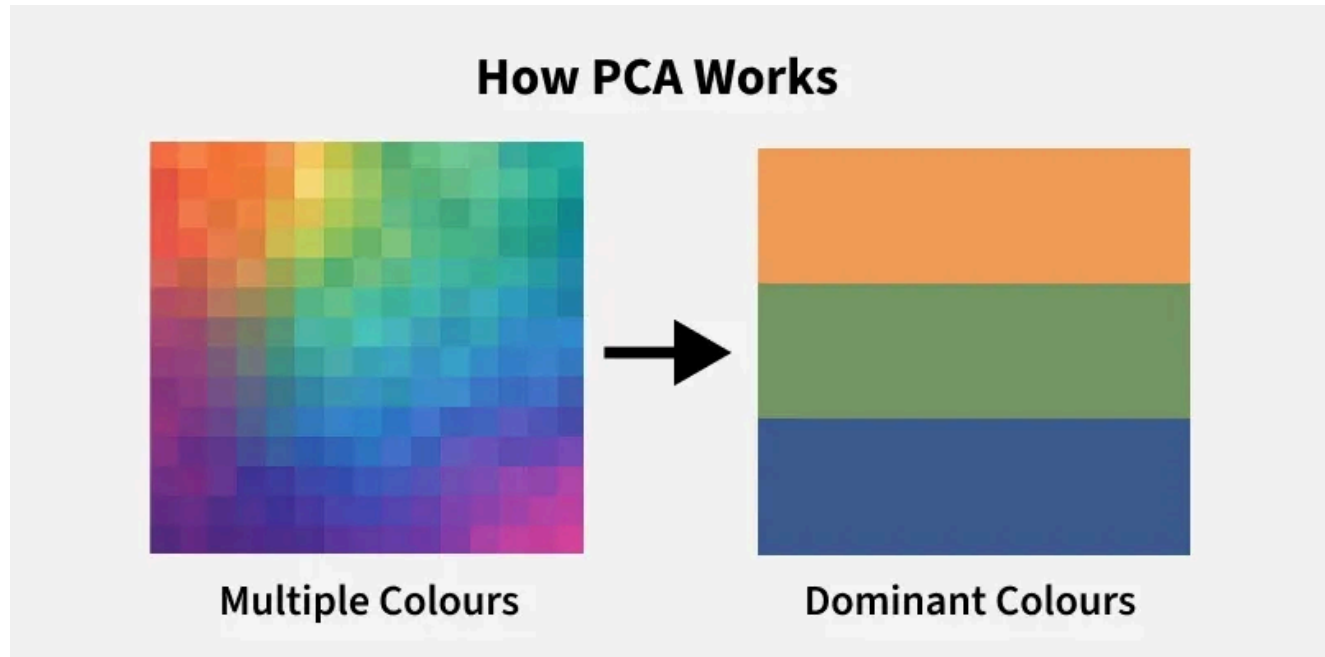


PCA (Principal Component Analysis)

PCA (Principal Component Analysis) is a **dimensionality reduction technique** and helps us to reduce the number of features in a dataset while keeping the most important information. It changes complex datasets by transforming correlated features into a smaller set of uncorrelated components.



It uses only **X** not **Y**, that make it **Unsupervised**.

PCA finding a **new axis** for your dataset.

So this become the new features called **Principle Component**

and each of this component is simply a combination of your previous feature.

Principal Component Requirement:

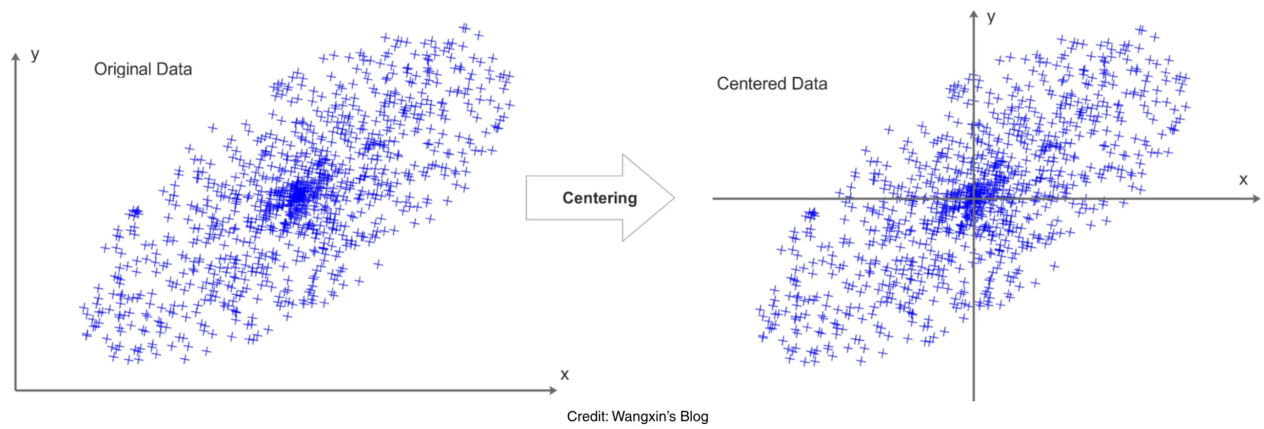
- Capture as much variance between the data point as possible
- Each principle component is completely uncorrelated with all the other components

PC1: Capture distinct pattern of variation

PC2: Capture distinct pattern of another variation

--> No overlap

Variance = the squared distance of data points from the mean



--> Maximizing sum of squared projection = Maximizing Variance captured along u

Eigenvector = Principal Components

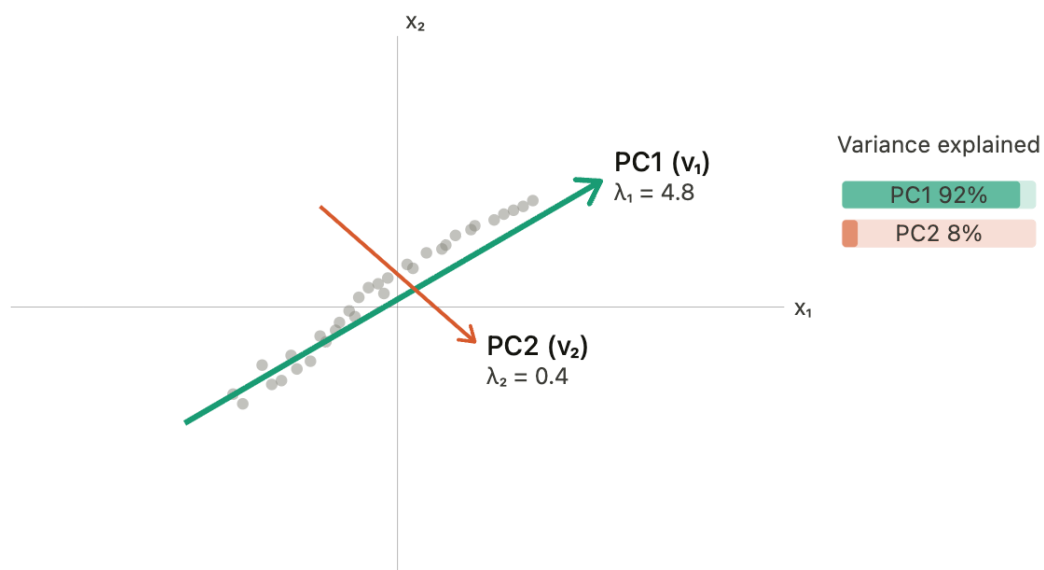
Eigenvalue = Explained variance of the component

Step 1: The Covariance Matrix

First, we compute the covariance matrix $\mathbf{C}$ from the data.

This matrix captures how every pair of features varies together. It is always square and symmetric.

Note: the Covariance Matrix is a linear transformation that describes the shape and spread of your data cloud. Its eigenvectors reveal the natural axes of that shape.



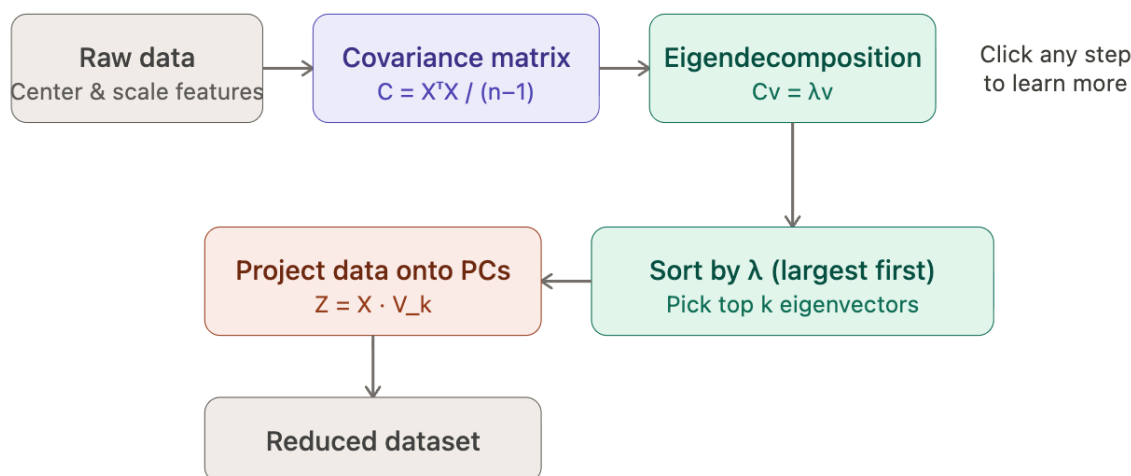
Step 2: Eigenvalues = Variance

Each eigenvalue λ tells you **how much variance** the data has along its corresponding eigenvector direction.

- A large $\lambda \rightarrow$ that direction spreads the data a lot $\rightarrow$ it's informative
- A small $\lambda \rightarrow$ that direction has little spread $\rightarrow$ it's mostly noise

In the diagram above, PC1 (the long axis) has $\lambda_1 = 4.8$, capturing 92% of the variance. PC2 (the short axis) has $\lambda_2 = 0.4$, only 8%. You could drop PC2 and lose very little information.

Step 3: The full PCA pipeline



$u_1 = \text{PC1}$

$u_2 = \text{PC2}$

as the vector is always symmetric, so the all the eigenvector are orthogonal to each other

All in all, the PCA essentially boils down to performing the Eigen decomposition of the covariance matrix of the original data.

PCA is not robust to outliers because it relies on the variance, which can be heavily influenced by the extreme values.

PCA looks for a low-dimensional representation of the observation that explains a good fraction of the variance.